Sahaj Bhadja

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Summary

Software engineer with 2+ years building cloud backend systems, REST APIs, and mobile-connected workflows. Experienced in TypeScript, Python, Java, AWS, microservices, authentication, concurrency, performance tuning, automated testing, DevOps, and production reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Reduced user-facing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling backend request paths and tuning Azure SQL queries, indexes, and secure access patterns.
- Built fault-tolerant asynchronous workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while sustaining 99%+ reliable production processing under load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask REST services and React workflows supporting 150+ users and 10,000+ daily requests, integrating secure ingestion, authentication, normalized AWS RDS persistence, status tracking, and usage instrumentation.
- Designed concurrent processing for 1,000+ uploaded files with real-time job-state updates, automated validation, and fault-tolerant execution, reducing turnaround 60% while maintaining stable API performance during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying structured automated tests, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a performance-critical architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Maintained 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving confidence in cross-platform production releases.
- Optimized distributed REST APIs using object-oriented patterns, filtering, pagination, indexing, and caching, improving response times 30% for bandwidth-constrained clients while increasing overall reliable production data-pipeline efficiency 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure user sessions across a platform handling 50,000+ records.
- Developed Flask REST APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and reliable webhook-driven synchronization that improved confirmation speed 40% across customer-facing native mobile application experiences.

Skills

Languages & Backend: TypeScript, JavaScript, Python, Java, C++, Flask, FastAPI, Node.js, REST APIs, JSON, object-oriented programming
User & Distributed Systems: authentication, secure sessions, microservices, WebSockets, concurrency, multithreading, asynchronous workflows, real-time synchronization, retries, failure isolation, scalability
Data & Cloud: AWS, Azure, Firebase, Firestore, AWS RDS, PostgreSQL, MySQL, SQL Server, Azure SQL, S3, Blob Storage, indexing, caching, latency tuning
Quality & DevOps: Git, CI/CD, Jenkins, GitHub Actions, Docker, PyTest, Mockito, unit testing, integration testing, observability, debugging, incident analysis, documentation, Agile, Scrum, production support

Projects

FinNexus | Live | *TypeScript, WebSockets, MongoDB, Kubernetes*
- Built a real-time financial platform supporting 50,000+ users and one million transactions, integrating secure bank linking, WebSocket synchronization, advanced filtering, and reliable scalable containerized production deployment.

HMIS | Live | *TypeScript, REST APIs, Appwrite, Twilio, Sentry*
- Built secure user authentication and consent-based API workflows with automated notifications and Sentry production observability, improving system reliability, debugging visibility, and responsiveness across customer-facing digital service experiences.

Big-Brain | Live | *Python, PostgreSQL, Clerk, GCP, REST APIs*
- Engineered secure document upload, user authentication, and cloud indexing workflows processing 50,000+ records, implementing reliable incremental updates and performance tuning to reduce retrieval latency 40% consistently.